

Class #6 - Discrete Dynamical Systems



1. Example: In a certain region, about 4% of the city's population moves to the surrounding suburbs each year and about 3% of the suburban population moves into the city. In 2026, there were 600,000 residents in the city and 400,000 in the suburbs. Set up a difference equation that describes this situation, where x_0 is the initial population in 2026. Then estimate the population in the city and in the suburb two years later, in 2028. (Ignore other factors that might influence the population size.)

2026

$$x_0 = \begin{bmatrix} C_0 \\ S_0 \end{bmatrix} = \begin{bmatrix} 600,000 \\ 400,000 \end{bmatrix}$$

Diagram showing population flow: City (C) to Suburbs (S) at 4%, and Suburbs (S) to City (C) at 3%.

2027

$$x_1 = \begin{bmatrix} C_1 \\ S_1 \end{bmatrix} = \begin{bmatrix} 0.96 C_0 + 0.03 S_0 \\ 0.04 C_0 + 0.97 S_0 \end{bmatrix}$$

2028

$$x_2 = \begin{bmatrix} C_2 \\ S_2 \end{bmatrix} = \begin{bmatrix} 0.96 C_1 + 0.03 S_1 \\ 0.04 C_1 + 0.97 S_1 \end{bmatrix}$$

$$x_k = \begin{bmatrix} C_k \\ S_k \end{bmatrix} = A \begin{bmatrix} C_{k-1} \\ S_{k-1} \end{bmatrix}$$

$$A = \begin{bmatrix} 0.96 & 0.03 \\ 0.04 & 0.97 \end{bmatrix}$$

Discrete Dynamical sys.

$k \rightarrow \infty$?

2. Example: A Predator-Prey System

In the redwood forests of California, wood rats provide up to 80% of the diet for the spotted owl, the main predator of the wood rat. Though unrealistic, a linear model can be set up as follows:

k : time in months

O_k : the number of owls

R_k : the number of rats (measured in thousands)

$R_0 = 5$

(5,000 rats)

(1) With no rats for food, only half of the owls will survive each month.

(2) With no owls as predators, the rat population will grow by 10% each month.

(3) If rats are plentiful, the owl population will rise by $0.4R_k$.

(4) Each month, $0.104O_k$ many of rats are eaten by owls.

$$x_k = \begin{bmatrix} O_k \\ R_k \end{bmatrix}$$

$$x_0 = \begin{bmatrix} O_0 \\ R_0 \end{bmatrix}$$

$$x_k = \begin{bmatrix} 0.5 & 0.4 \\ -0.104 & 1.1 \end{bmatrix} x_{k-1}$$

$k = 1, 2, 3, \dots$

$$x_1 = \begin{bmatrix} O_1 \\ R_1 \end{bmatrix} = \begin{bmatrix} 0.5 O_0 + 0.4 R_0 \\ R_0 + 0.1 R_0 - 0.104 O_0 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.4 \\ -0.104 & 1.1 \end{bmatrix} \begin{bmatrix} O_0 \\ R_0 \end{bmatrix}$$

If we get two distinct real eigenvalues λ_1 and λ_2 with $|\lambda_1| > 1$ and $|\lambda_2| < 1$, and eigenvectors $\vec{v}_1$ and $\vec{v}_2$ respectively, then

$\vec{x}_k = A \vec{x}_{k-1}$ A 2×2 $\vec{x}_k$ $\text{in } \mathbb{R}^2$ $\vec{x}_{k-1}$ $\text{in } \mathbb{R}^2$ $B = \begin{bmatrix} O_k \\ R_k \end{bmatrix}$ $B = \{ \vec{v}_1, \vec{v}_2 \}$ $k=1, 2, 3, \dots$

$$\vec{x}_0 = c_1 \vec{v}_1 + c_2 \vec{v}_2 \quad \text{for some } c_1, c_2$$

$$\begin{aligned} \vec{x}_1 &= A \vec{x}_0 = A (c_1 \vec{v}_1 + c_2 \vec{v}_2) \\ &= c_1 (A \vec{v}_1) + c_2 (A \vec{v}_2) \\ &= c_1 \lambda_1 \vec{v}_1 + c_2 \lambda_2 \vec{v}_2 \end{aligned}$$

$$\vec{x}_k = c_1 \lambda_1^k \vec{v}_1 + c_2 \lambda_2^k \vec{v}_2$$

$$\lim_{k \rightarrow \infty} \vec{x}_k = \lim_{k \rightarrow \infty} (c_1 \lambda_1^k \vec{v}_1 + c_2 \lambda_2^k \vec{v}_2)$$

$|\lambda_2| < 1$

$$\approx c_1 (\lambda_1)^k \vec{v}_1$$

$\uparrow$
fixed

For example, if the eigenvalues are 1.05 (with the corresponding eigenvector (6, 13)) and 0.55:
Both populations will grow at 5% per ~~year~~ ^{month}. Eventually, there will be approximately 6 spotted owls to every 13 (thousand) rats.

For k big

$$\lambda_1 = 1.05$$

$$\vec{v}_1 = \begin{bmatrix} 6 \\ 13 \end{bmatrix}$$

$$\lambda_2 = 0.55$$

not interested in $\vec{v}_2$
for large k

$$\begin{bmatrix} O_k \\ R_k \end{bmatrix} = \vec{x}_k \approx c_1 (1.05)^k \begin{bmatrix} 6 \\ 13 \end{bmatrix}$$

$$\begin{aligned} \begin{bmatrix} O_{k+1} \\ R_{k+1} \end{bmatrix} &= \vec{x}_{k+1} \approx c_1 (1.05)^{k+1} \begin{bmatrix} 6 \\ 13 \end{bmatrix} = (1.05) \left(c_1 (1.05)^k \begin{bmatrix} 6 \\ 13 \end{bmatrix} \right) \\ &= (1.05) \cdot \vec{x}_k \end{aligned}$$

$$\begin{aligned} &= (1.05) \begin{bmatrix} O_k \\ R_k \end{bmatrix} = \begin{bmatrix} O_k \\ R_k \end{bmatrix} + 0.05 \begin{bmatrix} O_k \\ R_k \end{bmatrix} \\ \frac{O_k}{R_k} &= \frac{c_1 (1.05)^k 6}{c_1 (1.05)^k 13} \end{aligned}$$

For example, if the eigenvalues are 0.9 and 0.7:

$$\vec{x}_k = c_1 (\lambda_1^k) \vec{v}_1 + c_2 (\lambda_2^k) \vec{v}_2$$

$\swarrow$ const. $\swarrow$ const.
 $\searrow$ 0 $\nearrow$ fixed $\searrow$ 0 $\nearrow$ fixed

$k \rightarrow \infty$
 $\vec{x}_k \rightarrow \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
 Both will vanish.

3. Example: Suppose the eigenvalues of a 3×3 matrix A are 3, $4/5$, and $3/5$, with corresponding eigenvectors $\begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \\ -5 \end{bmatrix}$, and $\begin{bmatrix} -2 \\ -5 \\ 3 \end{bmatrix}$. Let $x_0 = \begin{bmatrix} -2 \\ -5 \\ 3 \end{bmatrix}$. Find the solution of the equation $x_{k+1} = Ax_k$ for the specified x_0 , and describe what happens as $k \rightarrow \infty$.

$$x_0 = c_1 \vec{v}_1 + c_2 \vec{v}_2 + c_3 \vec{v}_3$$

$$B = \left\{ \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -5 \end{bmatrix}, \begin{bmatrix} -2 \\ -5 \\ 3 \end{bmatrix} \right\}$$

$$\vec{x}_{k+1} = c_1 (\lambda_1^k) \vec{v}_1 + c_2 (\lambda_2^k) \vec{v}_2 + c_3 (\lambda_3^k) \vec{v}_3$$

$$\vec{x}_{k+1} \approx c_1 (3^k) \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} \quad \text{For large } k.$$

$$x_0 = \begin{bmatrix} -2 \\ -5 \\ 3 \end{bmatrix}$$